Fraction Operations

Year 4 · **Mark Scheme**

Lesson Worksheet — Answers

Easy / Standard

1. 3/5

2. A — 5/7 (add the numerators: 3+2=5, keep the denominator)

3. 3/8

4. 4/6 (= 2/3 in simplest form)

5. B — 7/9

6. 4/6 (= 2/3)

7. 5/8

8. A and B (2/9+3/9 = 5/9 and 1/9+4/9 = 5/9; C and D both give 6/9)

9. 2/4 (= 1/2)

10. A — 4/5

11. 4/8 (= 1/2)

12. 5/10 (= 1/2)

13. 4/8 (= 1/2)

14. B — 3/7

15. 9/12 (= 3/4)

Standard

16. 6/8 (= 3/4)

17. A — 5/9

18. B — 5/6 (10/12 simplifies to 5/6)

19. 9/10

20. 2/7 (4/7 + 2/7 = 6/7)

21. 4/8 (= 1/2)

22. 5/10 m (= 1/2 m)

23. A and C (3/8−1/8 = 2/8 = 1/4 and 7/8−5/8 = 2/8 = 1/4; B = 3/8, D = 3/8)

24. 5/11

25. A — 8/9

26. 5/6 (already in simplest form)

27. 9/14

28. 4/12 litre (= 1/3 litre)

29. B — 4/5

30. 7/15 (8 − 3 + 2 = 7)

Standard / Hard

31. B — 1 3/8 (5/8 + 6/8 = 11/8 = 1 3/8)

32. 3/7 (3/7 + 3/7 = 6/7)

33. 12/5 (3 × 4 = 12, keep the denominator 5)

34. 3/4 (1/4 + 2/4)

35. C — 5/9 + 6/9 = 11/9, which is greater than 1 whole (the others are all less than 1 whole: 5/6, 6/7, 5/8)

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36. 5/9 (7 − 2 = 5, and 5/9 is already in its simplest form)

37. A and C (2/8 + 6/8 = 8/8 = 1 whole and 4/9 + 5/9 = 9/9 = 1 whole; B gives 5/7 and D gives 9/11, neither is 1 whole)

38. 9 − 3 = 6, so 6/10 left (= 3/5)

39. A — 4/7 (6/7 − 2/7 = 4/7, already in simplest form)

40. 2 3/4 (11 ÷ 4 = 2 remainder 3)

41. 8 − 3 = 5, so 5/10 m left (= 1/2 m)

42. C and D (5/8 + 5/8 = 10/8 and 5/8 + 6/8 = 11/8, both greater than 1 whole; 5/8 + 2/8 = 7/8 and 5/8 + 3/8 = 8/8 = 1 exactly, neither is greater than 1)

43. 3 × 4 = 12, so 12/9 (an improper fraction)

44. 9 − 4 = 5, so 5/10 left (= 1/2)

45. 9 − 4 = 5, so 5/10 m left (= 1/2 m)

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Homework Worksheet — Answers

Easy / Standard

1. 5/6

2. A — 7/9

3. 3/9 (= 1/3)

4. 5/8

5. A — 7/10

6. 5/11

7. 4/8 litre (= 1/2 litre)

8. A, B and D (2/7+4/7, 1/7+5/7 and 3/7+3/7 all equal 6/7; 3/7+2/7 = 5/7)

9. 11/12

10. A — 6/7

11. 9 − 1 − 3 = 5, so 5/10 (= 1/2)

12. 9/13

13. 5/9 m

14. B — 3/6 (= 1/2)

15. 3/9 (= 1/3)

Standard

16. 8/9

17. A — 4/11

18. B — 2/3 (6/12 + 2/12 = 8/12, which simplifies to 2/3)

19. 10/11

20. 4/8 (= 1/2)

21. 7/13

22. 4/10 kg (= 2/5 kg)

23. A, B and D (4/5−2/5, 3/5−1/5 and 5/5−3/5 all equal 2/5; 4/5−3/5 = 1/5)

24. 12/15 (= 4/5)

25. B — 3/7 (3/14 × 2 = 6/14, which simplifies to 3/7)

26. 5/6 + 1/6 = 6/6 (= 1 whole); 1 − 2/6 = 4/6 (= 2/3)

27. 5/16

28. 12 − 5 − 4 = 3, so 3/12 left (= 1/4)

29. B — 7/9

30. 9 − 2 + 3 = 10, so 10/14 (= 5/7)

Standard / Advanced

31. Half of 8/8 = 4/8; 4/8 + 1/8 = 5/8

32. A — 10/9 (1/9 + 4/9 = 5/9, doubled = 10/9)

33. Double 1/10 = 2/10; 6/10 − 2/10 = 4/10 (= 2/5)

34. A, B and D (double 3/9 = 6/9; 3/9 + 3/9 = 6/9; 3/9 × 2 = 6/9; only C, 2/9 + 3/9 = 5/9, is not equal)

35. 10 − 3 = 7, so 7/12 full

36. 8 − 2 − 1 = 5, so 5/10 kg left (= 1/2 kg)

37. C — 5/9 (8/9 − 3/9 = 5/9)

38. 10 − 3 = 7, so 7/12 m left

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39. 2/11 × 4 = 8/11

40. B and D (9 − 4 = 5, so 5/12 remains; this is not more than half, 6/12, so A is false; 5/12 is exactly 5/12, so B is true; 1/3 = 4/12, and 5/12 is not equal to that, so C is false; 5/12 is more than 4/12, so D is true)

41. 12 − 5 = 7, so 7/16 m left

42. B — 6/13 (10 − 4 = 6)

43. 12 − 5 = 7, so 7/15 m left

44. 8 − 3 = 5, so 5/10 (= 1/2)

45. Double 2/8 = 4/8; 4/8 − 1/8 = 3/8

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